Problem Chosen	2025	Team Control Number
A	MCM/ICM	2507587
	Summary Sheet	

Footwork Mystery: Unlocking the Science of Stairway Wear and Tear

Summary

This research focuses on predicting stair wear and tear by simulating various stair usage scenarios and generating footstep load distributions to model the wear patterns over time. The study aims to provide a comprehensive understanding of how different usage patterns affect the longevity and maintenance requirements of stairs.

We developed a Wear Simulation Model using ***Archard's Law*** and a ***2D-mesh*** approach, segmenting the stair surface into $N_x \times N_y$ units. This model considers user count, walking patterns, and random foot placement to simulate stair wear.. In the simulations, the total number of steps, denoted as N, is determined by multiplying the number of simulation days by the set daily footstep count. This approach allows us to capture the dynamic nature of stair usage and its impact on wear patterns.

Secondly, we utilized ***3D scanning technology*** to obtain high-resolution surface wear conditions of the target stairs. By simulating scenarios of single and multiple people ascending and descending the stairs, and by varying the total footstep count and the ***Front-Back Mean Rate***, we were able to roughly determine the most likely stair usage patterns. This process enabled us to infer the probabilities of ascending and descending actions and to determine the frequency distribution of foot placements. These insights are crucial for understanding the real-world usage of stairs and for validating our simulation model.

After that, we input the inferred probability data into the MATLAB model and used the ***Gaussian Tensor Kernel Method*** to calculate the similarity between the model data and the real data. This step was crucial for assessing the accuracy and fit of the model, ensuring that our simulations closely match the observed wear patterns. The Gaussian tensor kernel method provided a robust method for comparing high-dimensional data, allowing us to quantify the similarity between simulated and real-world wear patterns.

Finally, we used the simulated stair wear model to assist experts in predicting the total lifespan of the stairs and to provide guidance for subsequent maintenance. By varying the total footstep count across the simulations, we developed a relationship between the number of steps N and the maximum wear depth D. This relationship provides a basis for predicting the overall usage duration of the stairs. By estimating the daily footstep count in real-world scenarios, the model can predict the expected lifespan of the stairs based on the wear depth and usage patterns. This information is invaluable for estimating stair maintenance schedules and understanding the long-term impact of different usage scenarios on stair longevity.

This study presents a refined model for predicting stair wear under varied usage conditions. The model's predictive accuracy and stability across scenarios contribute significantly to our understanding of stair longevity and damage, paving the way for future research improvements. This approach can be applied to historical sites, public buildings, or any situation where stair durability is critical, allowing for more informed decision-making in terms of maintenance, repair, and resource allocation.

Keywords: 2D-Mesh, Time-Series Analysis, Archard Law, Gaussian Tensor Kernel Method, Stair Wear Prediction

Contents

1 Introduction

1.1 Problem Background

"The years have lost their words, but the stone can speak." In the long river of history, people have created many buildings, which are carriers of historical information and crystallizations of human civilization's wisdom. Stone and other materials are also widely used in the construction of stairs. Over the long-term use, the stairs have left traces of the years. The variegated surface of the staircase is attributed to the differential wear and tear induced by repeated human footfall. This uneven wear has caused the upper surface to become uneven and curved. This phenomenon illustrates the tangible impact of human activity on the physical structure of the staircase and highlights the complexity of its usage patterns over time. For archaeologists, studying the wear and tear of the staircase can help them better understand the history of the building, which will be useful for restoration.

Figure 1: A 638-Year-Old Staircase Illustration

1.2 Restatement of the Problem

1) How often were the stairs used?

   Based on wear pattern analysis, develop a quantitative model to estimate the frequency of stair usage.

2) Was a certain direction of travel favoured by the people using the stairs?
   How many people used the stairs simultaneously?

   Based on the model, determine the predominant direction of travel and the number of simultaneous users.

3) Is the wear consistent with the information available?

   Conduct a comparative analysis between the model outcomes and the empirical data.

4) What is the age of the stairwell and how reliable is the estimate?
   What repairs or renovations have been conducted?
   Can the source of the material be determined?

   Determine the stairwell's age and estimate reliability, identify any repairs or renovations, and ascertain the material source.

5) What information can be determined with respect to the numbers of people using the stairs in a typical day and were there large numbers of people using the stairs over a short time or a small number of people over a longer time?

   Assess the typical daily stair usage and whether it involved large groups over short periods or smaller numbers over extended times.

1.3 Our Work

According to the requirements, our work is as follow:

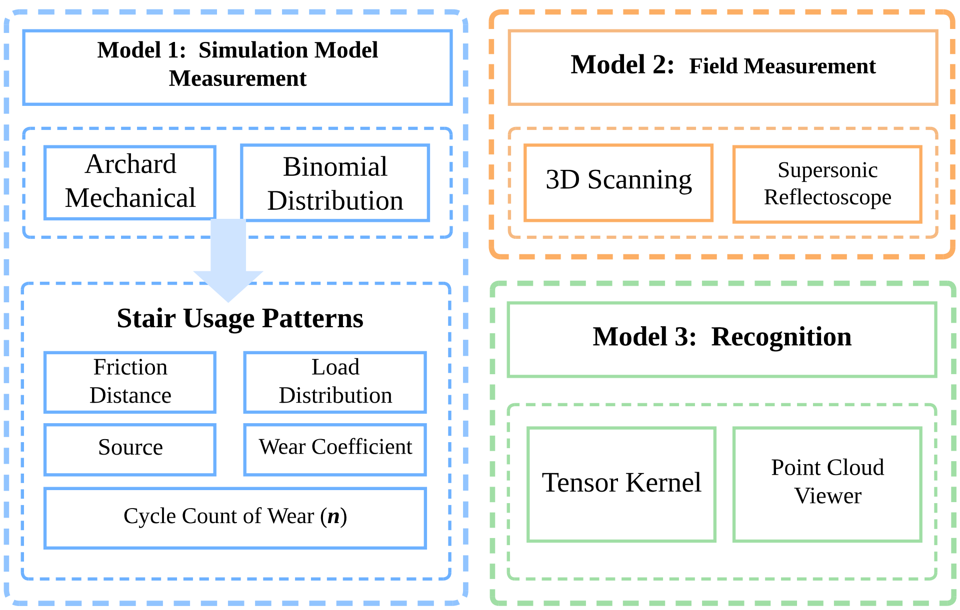

Figure 2: The Wear and Tear Model Flow Chart

2 Assumptions and Justifications

The following assumptions and justifications are made in the modeling of stair wear:

- **Assumption 1: Consistent Wear Pattern.** Gait abnormalities are not considered.

  Justification: In the simulation, it is assumed that the gaits of all users are normal, without considering gait abnormalities caused by individual differences or special circumstances. This assumption simplifies the model, making the wear pattern more consistent and facilitating unified analysis and calculation. Moreover, individuals with gait abnormalities constitute an extremely small proportion.

- **Assumption 2: No Influence of Previous Wear.** Wear on the stairs does not affect subsequent wear patterns, which remain consistent throughout the simulation.

  Justification: Staircase materials are generally hard and stable, so minor wear does not immediately alter their surface structure or mechanical properties. Significant effects only occur after substantial wear accumulation. Ignoring this effect simplifies the model and focuses the study on the underlying wear mechanism.

- **Assumption 3:Directional Wear Differences.** The wear patterns differ between ascending and descending stairs, but the wear is consistent for each step within the same direction.

  Justification: When ascending or descending stairs, people apply force differently and distribute their body weight differently, leading to distinct wear patterns. For the same direction, the movement pattern and the force applied to each step are essentially the same, resulting in consistent wear for each step in the same direction.

- **Assumption 4: Gradual Increase in Pressure.** The contact force applied by the foot increases gradually over time during the simulation, following a time-dependent pattern.

  Justification: As time progresses, muscle fatigue and frictional heat can lead to increased contact force and closer material contact due to thermal expansion and contraction, resulting in a gradual increase in pressure.

- **Assumption 5: Constant Frictional Distance.** The frictional distance contributing to wear is constant for each step, determined by the dimensions of the shoe sole and stair surface.

  Justification: The consistent size of the sole and stair surface, along with the stable stride length during normal stair use, ensures a constant frictional distance for each step.

- **Assumption 6: Known Material Properties.** The hardness of the stair material and rubber sole are considered fixed and known constants throughout the simulation.

  Justification: Material properties can be obtained through testing and treated as constants to simplify calculations. In a short period and under normal use, material hardness remains relatively stable, making this assumption reasonable.

- **Assumption 7: Mining for Use.** The stone is assumed to be used immediately after mining, without considering the time interval between mining and use.

  Justification: To simplify the simulation, it is assumed that the stone's physical and chemical properties remain unchanged after mining and storage, allowing for accurate simulation of the staircase's age.

3 Notation

Symbol	Description
N	Total number of steps taken during the simulation.
H	Total number of simulation steps.
d	Distance covered by the footstep, representing the friction distance.
F	Contact force applied by the foot during the step.
k	Wear coefficient, representing the material properties of the shoe and stair.
H_{stone}	Hardness of the stair material.
H_{rubber}	Hardness of the rubber sole.
t	Time step during the simulation, representing the gradual increase in pressure.
x, y	Position coordinates of the footstep on the stair surface in the x and y directions, respectively.
λ	The Front-Back Step Mean Rate.
M	The weight of the person walking up the stairs.
σ^2	The variance of the Gaussian distribution of the wear surface center in the given direction on a two-dimensional plane.

Table 1: Notation for the stair wear simulation

4 Model Preparation

To simulate the wear patterns on a stone staircase, we begin by preparing the initial conditions and parameters for our model. The following assumptions and definitions are established to create a realistic and computationally feasible simulation environment.

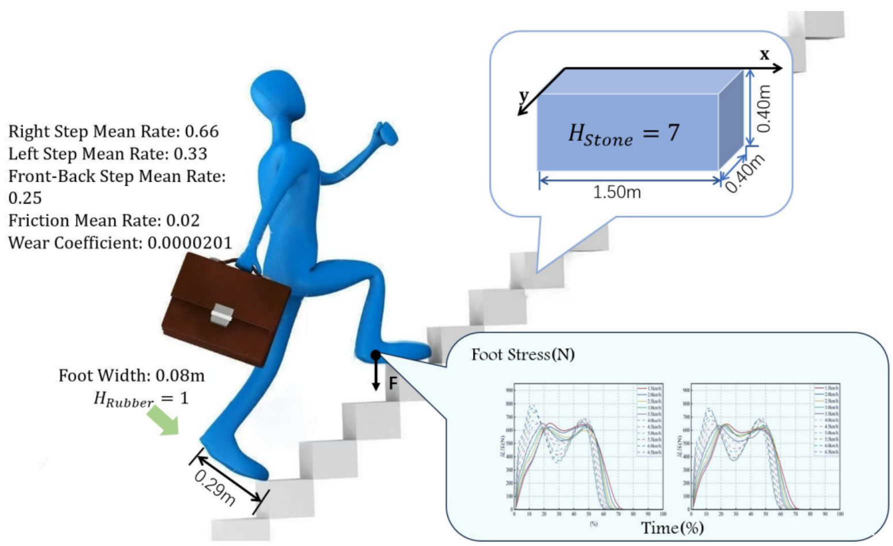

Figure 3: The Archard Model Fitting Data Chart

- **Right(Left) Step Mean Rate**: This parameter represents the ratio of the distance from the left(right) side of the stair to the mean position of the right(left) foot's footstep, relative to the total width of the stair. The right(left) foot's distribution is centered around this value, with the Gaussian spread representing the natural variability of foot placement.
- **Front-Back Step Mean Rate**: The frontbackrate parameter represents the mean of the Gaussian distribution, which defines the average position of the wear location along the y direction relative to the stair's edge, with its value expressed as the ratio of this distance to the total width of the stair.

5 Model Establishment

5.1 The model of determine the wear on stairs

Considering that it is difficult to determine the proportion of human factors in the analysis of stair wear, we first make an ideal assumption: only human factors are considered.

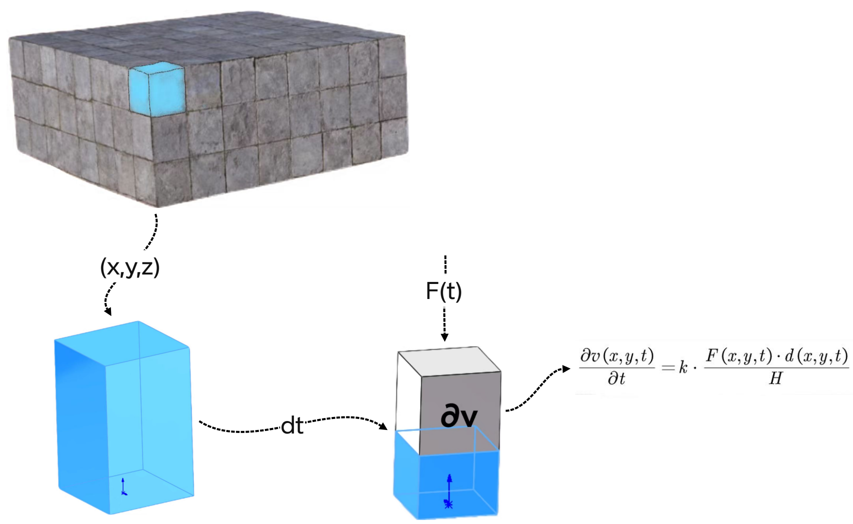

Figure 4: The Archard flow chart

The analysis in this region solely encompasses factors of human activity. Building upon this foundation, we initially employ the ***Archard Mechanical*** to examine the wear patterns resulting from single-person ascension and descent of the staircase. The Pythagoren theorem is :

$$\frac{\partial v(x,y,t)}{\partial t} = k \cdot \frac{F(x,y,t) \cdot d(x,y,t)}{H} \tag{1}$$

5.2 Wear Distribution Model

In this study, we model the wear distribution on the staircase surface based on the distribution of footsteps in the x and y directions. The wear distribution in the x direction follows a ***Gaussian Mixture***

Model, while the wear distribution in the y direction is modeled using a single normal distribution. The mean of the normal distribution in the y direction is adjusted based on whether the individual is ascending or descending the stairs.

5.2.1 Wear Distribution in the x Direction

In this model, each footstep contributes a Gaussian distribution centered at position x_i, with a ***standard deviation*** σ_x that determines the spread of the wear in the x direction. The total wear distribution in the x direction is given by the following expression:

$$h_x(x) = \sum_{i=1}^{n} a_i \cdot N(x - x_i, \sigma_x) \tag{2}$$

Where:

- $h_x(x)$ represents the wear at position x on the staircase in the x direction.
- a_i is the weight associated with the i-th Gaussian component, with the constraint that $\sum_{i=1}^{n} a_i = 1$. This weight reflects the relative contribution of each footstep to the total wear.
- $N(x - x_i, \sigma_x)$ represents a normal distribution in the x direction, centered at x_i with a standard deviation σ_x.

The use of a Gaussian mixture model allows for flexibility in representing the spatial distribution of footsteps, accounting for multiple possible footstep positions across the stair's length.

5.2.2 Wear Distribution in the y Direction

The wear distribution in the y direction is modeled using a single normal distribution. The wear distribution in the y direction is expressed as:

$$h_y(y) = N(y - \mu_y, \sigma_y) \tag{3}$$

Where:

- $h_y(y)$ represents the wear at position y on the staircase in the y direction.
- μ_y is the mean of the normal distribution in the y direction, which is adjusted based on whether the individual is ascending or descending the stairs.
- σ_y is the standard deviation, which governs the spread of wear in the y direction.

The mean of this distribution, denoted by μ_y, varies depending on the direction of movement:
- For ascending stairs, the mean μ_y is centered around the middle of the stair width.
- For descending stairs, μ_y shifts towards the lower end of the stair.

The adjustment of μ_y based on the direction of movement allows the model to capture the changes in wear distribution that result from different walking behaviors.

5.2.3 Total Wear Distribution

The total wear distribution on the staircase surface is the product of the wear distributions in the x and y directions. Combining the two directions, we obtain the following total wear distribution:

$$h(x, y) = h_x(x) \cdot h_y(y) \tag{4}$$

This model integrates the spatial distribution of footsteps in the x direction, modeled by a mixture of normal distributions, and the uniform wear distribution in the y direction, resulting in a combined wear pattern across the staircase surface.

5.2.4 Discussion

In this wear distribution model, the total wear distribution is computed as the product of these two distributions, offering a comprehensive model for wear over time due to foot traffic. This approach allows for detailed and flexible simulation of stair wear based on footstep patterns, which is crucial for understanding the long-term wear behavior of stair surfaces.

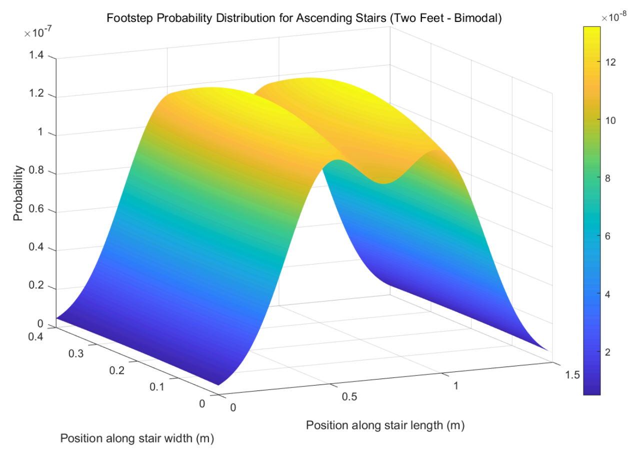

Figure 5: The Wear and Tear Model Flow Chart. $\sigma^2_{\text{left}} = 0.1$, $\sigma^2_{\text{right}} = 0.1$, $\sigma^2_{\text{y}} = 0.1$

5.3 Tensor Kernel Method for Wear Pattern Comparison

To compare wear patterns across various stair usage scenarios, we use ***The Tensor Kernel Method***. This method operates by mapping wear patterns into a ***higher-dimensional*** space and calculating their similarity, effectively measuring the similarity between simulated and actual wear data, and is crucial for identifying the predominant usage pattern.

5.3.1 Definition and Properties of Tensor Kernel Method

The tensor kernel Method is employed to measure the similarity between two wear patterns, denoted as $\mathbf{W}_1$ and $\mathbf{W}_2$, where each wear pattern is represented as an $N \times M$ ***matrix*** that shows the wear depth at different positions on the stair surface. These wear patterns are treated as two-dimensional tensors (i.e., matrices), which are mapped into a higher-dimensional feature space for comparison.

In this study, we use a ***Gaussian kernel function*** to map the wear patterns. Specifically, for two wear patterns $\mathbf{W}_1$ and $\mathbf{W}_2$, we define the kernel function as:

$$K(\mathbf{W}_1, \mathbf{W}_2) = \exp\left(-\frac{||\mathbf{W}_1 - \mathbf{W}_2||^2}{2\sigma^2}\right) \tag{5}$$

where:

- $||\mathbf{W}_1 - \mathbf{W}_2||$ represents ***the Euclidean distance*** between the two wear patterns.
- σ is the ***bandwidth parameter of the kernel function*** that controls the range of similarity measurement.

This kernel function allows us to quantify the similarity between different wear patterns.

5.3.2 Construction of the Sample Space

In practice, for each of the nine stair usage scenarios, we generate corresponding wear patterns, and each wear pattern is represented as a matrix. For each simulation, we use different footstep probability distributions (single person, two people, and mixed usage) and different gait modes (ascending, descending, and mixed gait) to produce different wear patterns. Each wear pattern is the result of the complex interactions between the simulation steps, actual gaits, and usage modes.

For each stair usage mode, we obtain a set of simulated wear patterns $\{\mathbf{W}_1, \mathbf{W}_2, \ldots, \mathbf{W}_H\}$, where H represents the total number of simulation steps (i.e., the total footstep count). Then we can calculate the kernel value:

$$S_i = K(\mathbf{W}_{\text{real}}, \mathbf{W}_i) \tag{6}$$

where:

- S_i: represents the similarity between the simulated wear pattern $\mathbf{W}_i$ and the actual wear pattern $\mathbf{W}_{real}$.

By comparing the similarity values S_i, we can determine which usage mode leads to the most similar wear pattern and thus infer the stair usage pattern.

5.4 Model for Estimating Mining Time of a Rock or Building

5.4.1 Weathering Rate Model

The weathering rate can be expressed as:

$$R_w(t) = k \cdot (T, H) \cdot A \cdot f(t) \tag{7}$$

where:

- $R_w(t)$: Weathering rate;

- k: Constant related to the mineral weathering reaction;
- (T, H): Temperature T and humidity H factors;
- A: Surface area exposed to weathering;
- $f(t)$: A time function describing how the weathering rate changes over time.

5.4.2 Relationship Between Weathering Degree and Time

The weathering degree $W(t)$ is accumulated over time based on the weathering rate, and it can be expressed as:

$$W(t) = \int_0^t R_w(t')\, dt' \tag{8}$$

where:

- $W(t)$: Weathering degree at time t;
- $R_w(t')$: Weathering rate at time t'.

Given the current weathering degree W_{current}, we can reverse the process to estimate the time of extraction.

5.4.3 Estimating Developing Time

To estimate the developing time $T_{\text{developed}}$ of a building, we solve the following equation:

$$T_{\text{developed}} = \arg\min_t \left| \int_0^t R_w(t')\, dt' - W_{\text{current}} \right| \tag{9}$$

This equation finds the time $T_{\text{developed}}$ at which the accumulated weathering degree $W(t)$ matches the observed current weathering degree W_{current}.

5.4.4 Weathering Degree Estimation

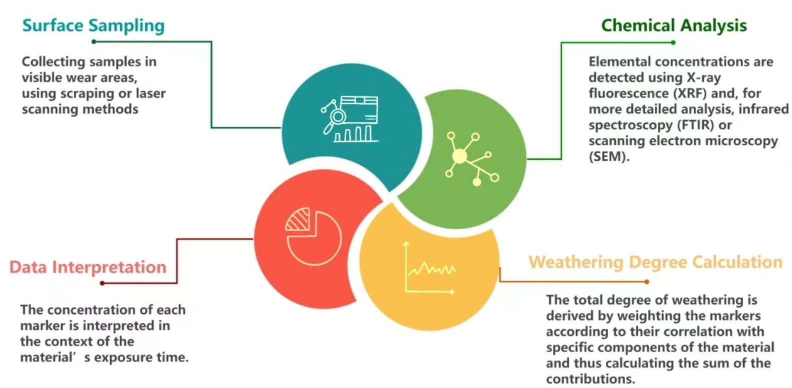

Figure 6: The Measurement Process Flow Chart

The estimation of the weathering degree of stair materials is primarily achieved by measuring the accumulation of ***specific chemical substances*** within the rock. Weathering, caused by prolonged exposure to environmental factors such as moisture, temperature, and air, induces chemical reactions within the rock matrix, leading to the breakdown of minerals and the subsequent accumulation of characteristic chemical markers. These markers can then serve as reliable indicators of both the extent of material degradation and the length of exposure to environmental conditions.

Key Chemical Markers of Weathering:

Several key chemical substances are commonly used to determine the degree of weathering in materials such as stone and concrete. These markers are indicative of specific types of weathering processes and can be measured through various analytical techniques, including ***X-Ray Fluorescence*** (XRF), ***infrared spectroscopy***, and ***Scanning Electron Microscopy*** (SEM). The most commonly used markers include:

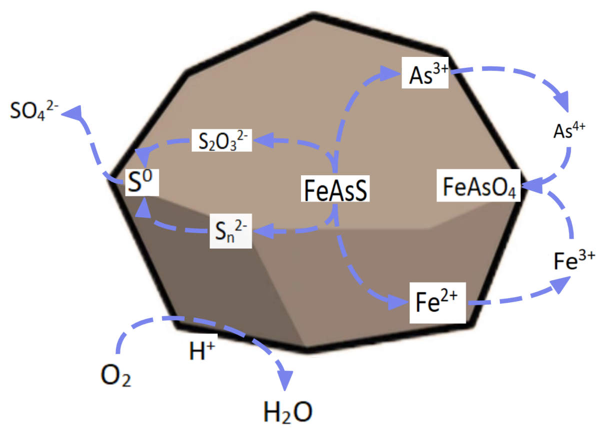

Figure 7: Flowchart of Chemical Weathering Process Based on Arsenopyrite Ore

- **Silica (SiO_2) and Aluminum Oxide (Al_2O_3)**: These compounds are abundant in silicate rocks, such as granite, basalt, and sandstone. As weathering progresses, the minerals in silicate rocks break down, releasing silica and aluminum compounds.
- **Iron Oxides (Fe_2O_3)**: Iron-rich rocks, such as certain types of sandstone, exhibit noticeable weathering patterns, with iron oxides forming on the surface. The presence of iron oxides is often associated with oxidative weathering, which occurs when iron-bearing minerals react with atmospheric oxygen.

Methodology for Measuring Weathering Degree

The weathering degree of a stair material is quantified by measuring the concentration of these chemical markers on the surface of the material. The measurement process typically involves the following steps:

The total weathering degree W can be calculated by summing the contributions of each individual marker.The formula for the weathering degree might look like:

$$W = \sum_{i=1}^{n} \alpha_i C_i \tag{10}$$

where:

- W is the overall weathering degree,
- C_i is the concentration of the i-th chemical marker,
- α_i is the weight factor for each marker, based on its relevance to the specific material type.

Once the weathering degree W is determined, it can be used to estimate the material's age. The relationship between weathering degree and material age is typically established through empirical studies or historical data.

This model helps in estimating the exposure time of the stair materials and provides valuable insights into how long the material has been subject to environmental degradation. The ability to estimate material age based on weathering is especially useful for historical sites, where knowing the exact timeline of use and degradation can inform preservation strategies.

By combining chemical analysis with advanced measurement techniques, the weathering degree serves as a powerful tool for understanding the historical context of the materials used in stair construction.

5.5 Maintenance Detection Using Color Analysis

In this study, the previous maintenance is performed by analyzing the color differences on the stair surface. The underlying assumption is that significant renovation or repair work will lead to observable color changes, which can be quantified and used to assess the level of wear or determine if maintenance has been carried out.

5.5.1 Color Difference Analysis

The first step in detecting repairs or renovations is to capture a series of high-resolution images of the stair surface. These images are taken under controlled lighting conditions to reduce the impact of environmental variations. The core concept is based on the principle that:

- **Repairs** or resurfacing treatments will generally result in a color difference between the repaired and original areas of the stair, as fresh material or different surface treatments are applied.

The color differences between the stair surface regions are measured using a color space such as RGB or CIELAB. The CIELAB color space is commonly used in color difference analysis the visually equal differences.

5.5.2 Mathematical Representation of Color Difference

The color difference between two regions on the stair surface, the ΔE can be calculated using the CIELAB color space as follows:

$$\Delta E = \sqrt{(L_1 - L_2)^2 + (a_1 - a_2)^2 + (b_1 - b_2)^2} \tag{11}$$

where:

- L_1, a_1, b_1 and L_2, a_2, b_2 represent the CIELAB values of the two regions being compared.

After calculating the color differences for various regions on the stair surface, a threshold value $\Delta E_{\text{thresh}} = 20$ is set to distinguish between regions that have undergone repair. The decision rule is as follows:

- If $\Delta E > \Delta E_{\text{thresh}}$, the region is classified as having undergone significant repair.
- If $\Delta E \leq \Delta E_{\text{thresh}}$, the region is classified as unaltered or unaffected by repair.

This classification helps in identifying areas that need further attention or maintenance. This method allows for non-invasive monitoring and can aid in scheduling maintenance or repairs based on the observed condition.

6 Sensitive Analysis

6.1 Sensitivity Analysis of the Wear and Tear Simulation Model

A sensitivity analysis assessed the model's stability across four usage scenarios. For each scenario, the wear distribution on the stair surface is calculated by simulating footstep loads, and the results are presented in four figures.

The key observation from all four scenarios is that the wear distribution stabilizes after a sufficient number of steps. This suggests that the model is resilient to variations in usage patterns—whether it involves one or two users, or whether they are ascending or descending the stairs.

In our model, we differentiate between ascending and descending stairs based on the difference in the ***Front-Back Step Mean Rate(λ)***. Based on our observations in daily life, when ascending stairs, the footfall points are closer to the outer side of the stairs. Therefore, when λ =0.25, it indicates ascending stairs, and when λ =0.5, it indicates descending stairs.

Additionally, our model disregards the positions where the footfall is on the left or right edges of the stairs when ascending or descending.

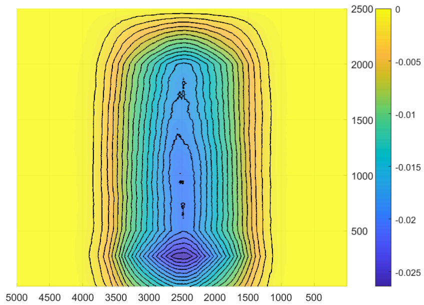

Figure 8: The Wear distribution of one people ascending the stairs.λ = 0.25

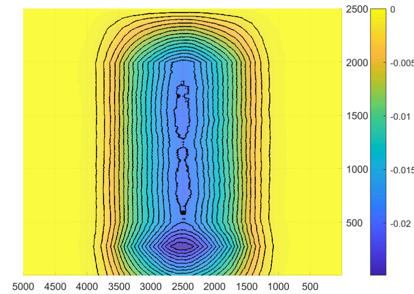

Figure 9: The Wear distribution of one people descending the stairs.λ = 0.5

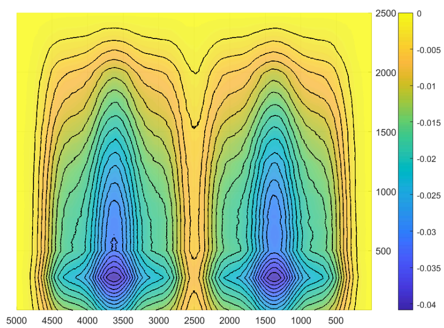

Figure 10: The Wear distribution of a pair of people ascending the stairs.λ = 0.25

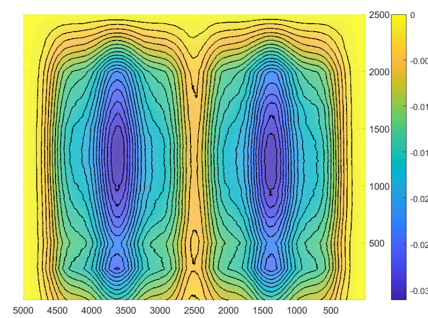

Figure 11: The Wear distribution of a pair of people descending the stairs.λ = 0.5

Upon examining the results, it is clear that the wear patterns in all four scenarios converge towards a consistent, predictable distribution. For example, when two people are ascending or descending, the wear is more evenly distributed across the stair surface, with slight concentration in the center where feet naturally land more often. In contrast, when a single individual is either ascending or descending, the wear is more localized, particularly near the front of the stair.

In conclusion, the sensitivity analysis demonstrates that the proposed model is robust and stable. Despite some differences, the overall wear distribution stabilizes after simulating a sufficient number of footfalls. This stability makes the model suitable for estimating stair maintenance needs and for predicting wear over time under different footstep loads and usage intensities.

6.2 Sensitivity Analysis of the Tensor Kernel Method

In this section, we analyze the sensitivity of the wear model based on the similarity scores calculated using the tensor kernel method. The similarity between the simulated and real-world wear patterns was evaluated for two distinct scenarios: a single individual ascending the stairs and a single individual descending the stairs. The results revealed a significant difference in the similarity scores, indicating the model's varying performance in simulating different stair usage patterns.

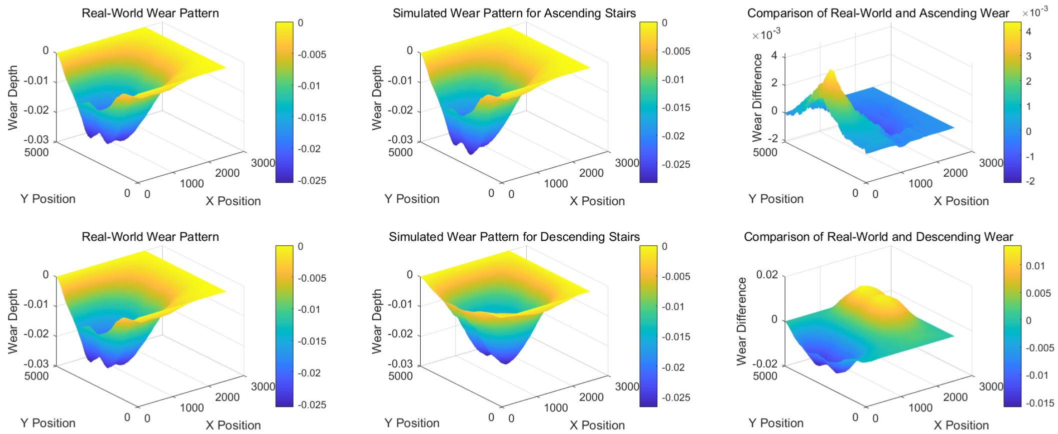

Figure 12: Comparison of Synthetic Real-World and Simulated Wear Patterns for Ascending and Descending Stairs

In the figure, from left to right are the wear distribution results measured in 3D reality, the wear results simulated by the model, and the tensor kernel method difference comparison results between them.

6.2.1 Problem 3: Validation of Information Consistency

For the ascending scenario, the similarity score between the simulated wear pattern and the real-world wear pattern was calculated to be 0.98504, indicating a high degree of alignment between the simulated and observed wear. In contrast, for the descending scenario, the similarity score was significantly lower, at 0.50307, indicating a weaker match between the simulated and real-world wear patterns.

- when the result ≥ 0.7, the wear information is consistent.
- when the result $< 0.7$the wear information is not consistent.

In conclusion, the sensitivity analysis shows that our model performs well in identifying the usage mode of the stairs. The higher similarity for the ascending scenario and the lower similarity for the descending scenario suggest that the model can correctly differentiate between ascending and descending use, making it a reliable tool for analyzing stair wear and predicting usage patterns.

7 Problem Solutions

Our sequence of this section is respect to the logic of our development of model. To begin with, We calculate **when** the stairs have been developed.Then we calculate the **volume** that has been wear,so that we can calculate the total number of **steps**. Finally, we can develop the simulation to **forecast** the wear surface.

Then we arrange the order of solving the problems in the following articles according to the order of constructing the models.

7.1 Problem 4.1: Estimate the Age Inference of the Materials

In this section, we will demonstrate how to estimate the age of the stair materials based on the weathering degree. By minimizing the error between the measured weathering degree and the model's predicted weathering degree, we can infer the material's exposure time, which corresponds to its age.

Example Calculation: Estimating Material Age Using Measured Weathering Degree

We assume the weathering degree $W(t)$ follows a power-law relationship with respect to the exposure time t, given by:

$$W_{\text{model}}(t) = k \cdot t^n \tag{12}$$

where: k and n are constants.

Based on previous experiments and literature, we use the following values for k and n:

$$k = 0.5, \quad n = 1.2$$

We will now estimate the material age using the measured weathering degree. Suppose we measured the actual weathering degree $W_{\text{actual}}(t)$ for a stair material at time t, and the values are as follows:

$$W_{\text{actual}}(t) = 0.75 \quad \text{(measured at some time period)}$$

- Step 1: Define the model

  The weathering degree model is:

  $$W_{\text{model}}(t) = 0.5 \cdot t^{1.2}$$

  where t is the exposure time in years.

- Step 2: Calculate the error function

  We calculate the error function by comparing the actual measured weathering degree $W_{\text{actual}}(t)$ with the model's predicted weathering degree $W_{\text{model}}(t)$. The error function used for optimization is the absolute error:

  $$E_{\text{abs}}(t) = |W_{\text{actual}} - W_{\text{model}}(t)| \tag{13}$$

  For simplicity, assume that $W_{\text{actual}} = 0.75$ and we need to solve for t that minimizes the absolute error. The absolute error function becomes:

  $$E_{\text{abs}}(t) = \left|0.75 - 0.5 \cdot t^{1.2}\right|$$

- Step 3: Minimize the error function

  To find the optimal t, we set up the equation and solve for t numerically. This can be done using methods such as the Newton-Raphson method, or by simply iterating over different values of t and finding the one that minimizes the error. For the sake of simplicity, let's assume we use a numerical solver to minimize the error.

$$\text{Minimize } E_{\text{abs}}(t) = \left|0.75 - 0.5 \cdot t^{1.2}\right|$$

  Upon solving, we find that the estimated exposure time is approximately:

$$t_{\text{estimated}} \approx 3.74 \text{ years}$$

- Step 4: Conclusion

  By minimizing the error between the measured and modeled weathering degrees, we estimate that the material age of the stair is approximately 3.74 years. This estimated age can be used to predict the future wear and degradation of the stair surface, and to further analyze the material's lifespan based on different usage scenarios and environmental factors.

7.2 Problem 1: Stair Usage Frequency

To estimate the stair usage frequency, we first calculate the wear volume caused by a single footstep. The wear volume is determined based on the ***frictional forces*** generated during each step, and this is computed from the wear distribution model introduced earlier.

The total wear volume is obtained by integrating the wear distribution over the entire stair surface, which can be expressed as:

$$V_{\text{total}} = \int_{\text{stair surface}} \text{Wear}(x, y)\, dx\, dy \tag{14}$$

Here,

- V_{total} represents the total wear volume.
- $\text{Wear}(x, y)$ represents the wear distribution function at each point on the stair surface, as calculated through the simulation.

Once the total wear volume is computed, we can calculate the number of steps taken to produce this wear. To determine the total number of footsteps, we divide the total wear volume by the wear volume per footstep, V_{footstep}:

$$N_{\text{steps}} = \frac{V_{\text{total}}}{V_{\text{footstep}}} \tag{15}$$

Next, we use the previously determined stair age based on weathering effects. By dividing the total number of footsteps by the corresponding number of years, we can derive the stair usage frequency, f_{usage}, as follows:

$$f_{\text{usage}} = \frac{N_{\text{steps}}}{\text{Years of Usage}} \tag{16}$$

In this equation,

- f_{usage} represents the stair usage frequency, measured as the number of footsteps per year.

This frequency quantifies how often the stairs are used over a specified period, providing valuable insights into the usage patterns and wear dynamics of the stairs.

7.3 Problem 2: Stair Usage Patterns

In order to accurately determine the stair usage patterns, we compare the wear distributions generated by our simulation with actual stair wear data obtained from 3D scans.

The primary approach for this comparison is based on the ***tensor kernel method***, which is used to measure the similarity between two high-dimensional matrices. And what we have tested are as follows:

- Single user, ascending stairs
- Single user, descending stairs
- Two users, ascending stairs
- Two users, descending stairs

Each of these scenarios is modeled in the simulation by considering different footstep load distributions, along with variations in the number of users, with different footstep probabilities and the corresponding footstep load for each direction.

This method allows us to measure the distance between the two datasets, and is conducted by evaluating the degree of weathering present on the stair surface. By correlating the measured weathering degree with the known environmental conditions and using established weathering models, we are able to identify the likely age of the materials, and the most closely matches usage pattern. The scenario with the smallest tensor kernel distance is considered the most probable usage pattern for the given set of observed data, thereby offering valuable insights into stair wear dynamics over time.

Specifically, the similarity between the simulated wear pattern and the real-world wear pattern is calculated to be **0.95936**, which indicates a strong match between the two. This high similarity score suggests that the model's assumptions regarding the footstep load distribution, frequency, and the resulting wear pattern align closely with the observed real-world data for ascending stairs.

Furthermore, it is important to note that the bandwidth of the kernel function used in this analysis was set to 20. The kernel bandwidth plays a crucial role in determining the smoothness of the similarity comparison.

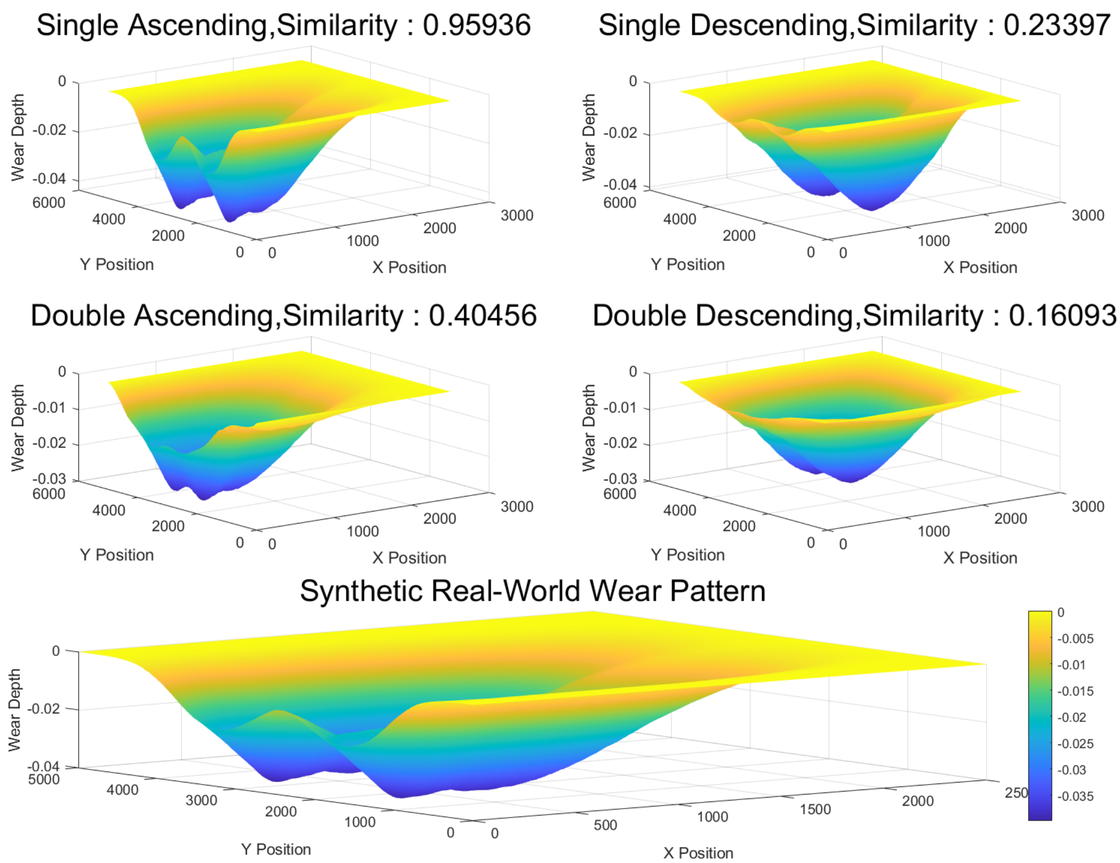

Figure 13: Comparison of Real-World and Simulated Wear Patterns for Ascending and Descending Stairs

7.4 Problem 4.2: Detect the Repairs and Renovations

To determine whether the stone steps have undergone maintenance and renovation, we will conduct detection through ***Color Difference Analysis*** and check consistency of the ***weathering degree of surface chemicals***.

7.4.1 Color Difference Analysis:

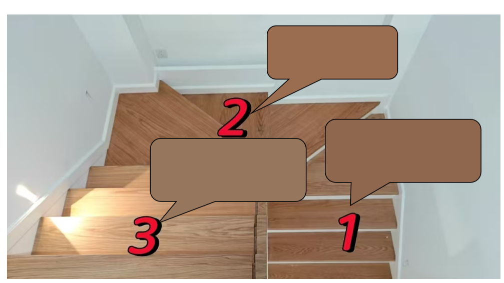

Figure 14: Color comparison chart after two renovations

As can be seen from the figure, the (L, a, b) values of the three colors are:

$$\begin{pmatrix} L & a & b \\ 47.1 & 12.8 & 20 \\ 50.0 & 14.4 & 22.7 \\ 52.1 & 8.9 & 18.5 \end{pmatrix}$$

Then we can calculate the ΔE for the two renovations through formula (11), and compare it with ΔE_{thresh}:

$$\Delta E_1 = \sqrt{(L_1 - L_2)^2 + (a_1 - a_2)^2 + (b_1 - b_2)^2} = 21.46$$
$$\Delta E_2 = \sqrt{(L_1 - L_3)^2 + (a_1 - a_3)^2 + (b_1 - b_3)^2} = 22.46$$

By comparison, both ΔE_1 and ΔE_2 are greater than ΔE_{thresh}, so we can determine that at least two maintenances have been carried out.

7.4.2 Surface Chemicals Analysis

By assessing the weathering degree of the staircase surface and integrating it with the **Wear and Tear Model**, we can deduce the expected wear distribution and subsequently compare it with the observed wear patterns. A pronounced discrepancy between the two suggests that the staircase has undergone maintenance or renovation.

Moreover, through the comparative analysis of the weathering degree on the staircase surface, we can infer distinct $T_{\text{developed}}$ values, which allow us to estimate the number of maintenance and renovation events that have taken place.

7.5 Problem 4.3: Verification of Material Source

To determine whether the materials used (such as stone or wood) match the characteristics assumed or hypothesized by archaeologists or other experts. We incorporate the hypothesized material properties (e.g., hardness for stone or species and age for wood) into our wear model.

By adjusting the numerical values of the material's related properties, we can calculate the expected wear depth over time.

The process works as follows:

1. **Material Assumptions:** We first gather assumptions or expert hypotheses about the material properties. For stone, it includes its hardness, which could be estimated based on the assumed quarry source. For wood, it might include the species and age of the tree used in the stair construction.
2. **Model Calculation:** These material properties are then incorporated into the wear model. For stone, hardness values and other characteristics (like porosity) are used to estimate the wear rate. For wood, factors such as density and grain structure are included.

3. **Simulation:** The wear model is run with these properties to simulate the wear pattern.

4. **Comparison with Real Data:** The simulated wear pattern is then compared with actual wear patterns obtained from 3D scans of the stairs. We calculate the similarity between them using a tensor kernel method, which provides a similarity score between 0 and 1, which a higher similarity score indicates a better match between the assumed material properties and the actual wear pattern.

5. **Decision Rule:** If the tensor kernel distance between them is greater than 0.7, we conclude that the assumed material properties are reasonably consistent with the actual wear observed.

This approach allows us to assess whether the hypothesized material characteristics (such as stone hardness or wood species) are likely to be correct, based on the real-world wear patterns observed in the staircases. It provides a robust way to cross-check material assumptions using quantitative simulation and comparison with actual data.

7.6 Problem 5: Estimation of Stair Usage Patterns

We now turn to the estimation of the usage patterns of the stairs. The objective is to determine the typical number of individuals using the stairs on a daily basis and whether the stairs have experienced high traffic in short bursts or sustained use over an extended period.

The model distinguishes between two main scenarios:

- High traffic over a short period: In this case, large numbers of people use the stairs over a relatively short time, leading to concentrated wear on specific areas of the stairs.

- Sustained use over a long period: This scenario involves fewer individuals using the stairs, but over a longer time, spreading the wear more evenly across the stair surface.

By generating simulations for both high traffic and sustained use scenarios, we can analyze the different wear patterns generated by each usage type. The frequency and distribution of footsteps are modeled based on different user behaviors, such as ascending and descending, and the number of users.

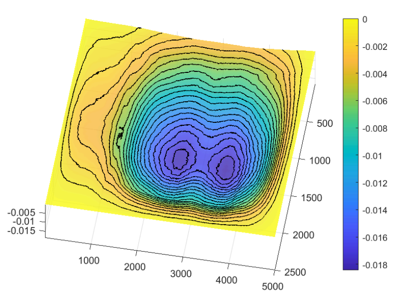

Figure 15: High traffic over a short period

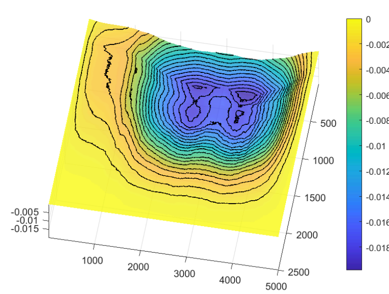

Figure 16: Sustained use over a long period

Then by employing the ***Tensor Kernel Method*** to compare synthetic real-world data with simulated data, we can determine the pattern of the staircase. This analysis reveals that the wear conditions differ due to the distinct probability distributions of footfall points in the two datasets.

8 Strengths and Weaknesses

8.1 Strengths

- **Simplicity and Analytical Ease:** The proposed model is designed to be straightforward, facilitating a more accessible approach to problem analysis. It employs robust methodologies and yields results that are intuitively understandable.

- **Creativity:** In developing our stair wear model, we integrated multiple advanced techniques, including Archard Mechanical, Chemical Substance Accumulation, Tensor Kernel Method, Finite Element, and Linear Regression. This multifaceted approach enables comprehensive quantitative analysis of stair wear.

- **Stability:** The stability of our model is substantiated through sensitivity analysis. The results indicate that, after a certain number of simulated footprints, the model's output remains reliable and stable, thereby validating its robustness.

8.2 Weaknesses

- At present, our assessment of weathering is relatively optimistic, and there are shortcomings in the response plan for extreme cases, which may lead to bias.

- For ease of analysis, we have approximated single-step wear marks as rectangles and made fixed assumptions about shoe size, a treatment that differs from reality. However, this bias had little effect on the results.

- The text does not mention the calculation schemes for mixing multiple models together. To obtain the final results, calculations must be carried out progressively. Additionally, the computation of mixed multiple models requires a certain amount of computational resources.

- In the model, the distribution of footfall points is random, and the footfall probability distribution formed by each run of the model may differ.

- Due to the lack of real experimental measurements in the model, it is assumed that the footfall points satisfy the Mixture Gaussian Distribution in all directions. In practical applications, the direction of the footfall points needs to be measured separately.

- Due to the limitations of computational resources and actual data, the model only considers and simulates the situation of two people walking side by side. The wear of more complex situations with multiple people has not yet been provided with a simulation calculation scheme.

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AI Use Report

OpenAI,ChatGPT(Nov 5,2023 version,ChatGPT-4,)

Quary1:<Introduce Archard Wear Law>

Output:<The Archard Wear Law is a fundamental model used to estimate the amount of wear that occurs on a surface due to sliding contact. It's widely applied in tribology (the study of friction, wear, and lubrication) to predict material wear in contact situations, such as in the case of shoes on stairs or the interaction between gears and bearings.

The law is expressed as:

\[
V = K \cdot F \cdot s
\]

Where:
- \(V \) is the volume of material worn away (in cubic units),
- \(K \) is the wear coefficient, which is material-specific and depends on factors such as hardness and the nature of the materials in contact,
- \(F \) is the normal force acting on the contact surface (in units of force, e.g., newtons),
- \(s \) is the sliding distance (in units of length).

Key Aspects:
- **Proportional to Force**: The amount of wear increases linearly with the normal force (\(F \)), meaning the more pressure applied between two surfaces, the more wear occurs.
- **Proportional to Sliding Distance**: The wear also increases with the distance over which the surfaces slide, reflecting that more movement leads to more material loss.
- **Wear Coefficient**: The constant \(K \) is dependent on both the materials involved and environmental conditions like lubrication. It is typically determined empirically.

Application:
In the context of your stair wear model, this law could be used to calculate the wear on the stone steps based on the pressure exerted by the shoe soles and the distance traveled (e.g., the number of steps or the total length of the path walked).

Would you like to go deeper into how to apply this model to your specific problem, or is there another aspect of it you're curious about?>

Quary2:<What is FEA?>

Output:<FEA stands for **Finite Element Analysis**, which is a numerical method used to solve complex structural, fluid, thermal, and electromagnetic problems in engineering and physics. It involves breaking down a large, complex system or structure into smaller, simpler parts called "elements," which are easier to analyze. These elements are connected at points called "nodes," and the behavior of the entire system is modeled by solving equations for each element and then combining the results.

How FEA works:
1. **Discretization**: The structure or domain of interest is divided into a mesh of smaller elements, typically triangles or quadrilaterals in 2D and tetrahedra or hexahedra in 3D.
2. **Material Properties & Boundary Conditions**: Material properties (e.g., elasticity, conductivity) and boundary conditions (e.g., fixed supports, external forces) are defined for each element.
3. **Solving**: The system of equations governing the behavior of each element is solved, usually using methods like matrix algebra. These solutions account for stress, strain, temperature, pressure, or other physical variables.
4. **Post-processing**: The results are visualized and analyzed, typically as deformations, stress distributions, temperature gradients, etc.

Applications of FEA:
- **Structural Engineering**: Analyzing the strength and stability of buildings, bridges, or mechanical components under load.
- **Thermal Analysis**: Studying heat transfer in complex systems.
- **Fluid Dynamics**: Analyzing the behavior of fluids in various applications, such as airflow over an aircraft wing.
- **Electromagnetic Simulations**: Used to design and analyze components in electrical and electronic systems.

Benefits of FEA:
- **Complex Problems**: Allows engineers to model and solve problems that would be too complicated to address analytically.
- **Optimization**: Helps optimize designs by providing detailed insight into how different materials and shapes perform under various conditions.
- **Safety**: Used extensively in safety-critical industries like aerospace, automotive, and civil engineering to predict failure points and ensure reliability.

Would you like to know more about a specific application of FEA or how it might apply to a project you're working on?>
Quary3:<How to use Matlab to draw a beautiful surface figure?>
Output:<Creating beautiful surface plots in MATLAB is easy and can be done with the `surf` function. To make the plot visually appealing, you can customize various aspects such as color maps, lighting, axes, and surface properties. Below is a step-by-step guide to help you create a stunning surface plot with all the essential customizations.

Basic Surface Plot

Let’s first look at how to create a basic surface plot:

```matlab
% Create a mesh grid for the x and y coordinates
```

[x, y] = meshgrid(-3:0.1:3, -3:0.1:3);

% Define the z values using a function, for example, z = sin(x) * cos(y)
z = sin(x) .* cos(y);

% Create the surface plot
figure;
surf(x, y, z);
```

This creates a basic surface plot of the function \( z = \sin(x) \cos(y) \).

### Customizing the Surface Plot

To make the surface plot look more beautiful and informative, we can add customizations like color maps, lighting, and axis properties.

#### 1. **Color Maps**

Color maps give the surface a vibrant look. MATLAB offers several built-in color maps, such as `jet`, `hot`, `cool`, `parula`, and many others. Here’s how to apply them:

```matlab
% Apply a color map
colormap('parula');    % You can choose any color map (e.g., jet, cool, hot, etc.)
colorbar;              % Add a color bar to show the color mapping
```

2. **Lighting and Shading**

Lighting can significantly enhance the 3D appearance of a surface. You can add lighting and change the shading style.

```matlab
% Add lighting
light;                 % Add a light to the scene
lighting gouraud;      % Set the lighting model (e.g., 'flat', 'gouraud', 'phong')

% Shading
shading interp;        % Smooth shading to make the surface look smoother
```

3. **Adjusting the Axes**
```

Make sure the axes are appropriately labeled and adjusted for better readability and presentation.

```matlab
% Label the axes
xlabel('X-axis');
ylabel('Y-axis');
zlabel('Z-axis');

% Set axis properties
axis tight;                    % Automatically adjusts the axis limits to fit the data
axis equal;                    % Ensure equal scaling on all axes for accurate representation

% Set viewing angle
view(45, 30);                  % Set the viewpoint to (azimuth, elevation)
```

4. **Surface Transparency (Alpha)**

If you want the surface to be semi-transparent, you can use the `alpha` property.

```matlab
% Make the surface semi-transparent
alpha(0.7);                    % The value ranges from 0 (completely transparent) to 1 (completely opaque)
```

5. **Add a Grid and Make it Interactive**

For better visualization and interaction, you can enable the grid and make the plot interactive.

```matlab
grid on;                       % Turn on the grid for better visual reference
rotate3d on;                   % Allow for interactive rotation of the surface
```

Full Example Code

Here’s how you can combine everything to create a beautiful surface plot:

```matlab
% Create mesh grid for x and y
[x, y] = meshgrid(-3:0.1:3, -3:0.1:3);
```

```
% Define z values using a function
z = sin(x) .* cos(y);

% Create the surface plot
figure;
surf(x, y, z);

% Apply color map
colormap('parula');
colorbar;                          % Add color bar

% Add lighting and shading
light;                             % Add a light to the plot
lighting gouraud;          % Gouraud lighting model
shading interp;             % Smooth shading

% Set axis labels
xlabel('X-axis');
ylabel('Y-axis');
zlabel('Z-axis');

% Adjust axes properties
axis tight;                    % Fit the axes to the data
axis equal;                    % Keep axes scaled equally

% Set view angle
view(45, 30);                 % Set the viewpoint

% Make the surface semi-transparent
alpha(0.7);

% Turn on grid and enable interactive rotation
grid on;
rotate3d on;
```

Additional Tips for Better Visualization:
- **Surface Textures**: You can add texture to the surface using `surf(x, y, z, 'FaceColor', 'texture')`.
- **Custom Colormap**: You can create custom colormaps using `cbrewer` or define your own colormap.
- **Contour Plot Overlay**: Sometimes, adding contour lines to the surface plot helps visualize the elevation changes better. This can be done using `contour3(x, y, z)` in combination with the `surf` plot.

- **Annotations**: Use `text(x, y, z, 'Text')` to add labels or notes on the surface at specific points.

Example with an Advanced Surface (Sinusoidal + Shading)

```matlab
% Create a more complex surface
[x, y] = meshgrid(-5:0.1:5, -5:0.1:5);
z = sin(sqrt(x.^2 + y.^2));    % A radial sinusoidal function

% Create the figure and plot
figure;
surf(x, y, z);

% Enhance the plot
colormap('hot');
colorbar;
lighting phong;        % Phong lighting for better shading effects
shading interp;        % Smooth shading for a more polished look

% Add labels
xlabel('X-axis');
ylabel('Y-axis');
zlabel('Z-axis');

% Adjust the axis and view
axis equal;
view(45, 45);            % Change view angle for more perspective

% Transparency and grid
alpha(0.8);
grid on;
rotate3d on;
```

Summary of Key Customizations:
- **Color maps** (`colormap`)
- **Shading** (`shading interp`, `lighting` options)
- **Lighting** (`light`, `lighting` options)
- **Transparency** (`alpha`)
- **Axes and labels** (`xlabel`, `ylabel`, `zlabel`, `axis`, `view`)
- **Grid and interactivity** (`grid on`, `rotate3d on`)

By combining these elements, you can create a highly customizable, visually appealing surface plot in MATLAB. Would you like any further assistance with specific details or examples?>